

- *nmap*
 - This command uses diverse methods for scanning the ports and services, and for running network IP addresses.
- *netstat*
 - Shows the process state of the ports in the host.
 - Is able to identify only the authorised services that are running, and the connections made from different hosts.
- *nslookup*
 - Returns the query name records for the given domain name or IP address using the DNS resolver.
- *netcat*
 - Enables reading and writing data between hosts.
 - One of the most powerful commands, it is considered a Swiss army knife of networking tools.
- *dnsenum*
 - Returns as much information and all the IP addresses that are in use for a domain.
- *scanless*
 - Scans all the services and ports that are running and open in a host.
- *curl*
 - Performs data transfer over many protocols, commonly used for HTTP.
 - Other supported protocols are FTP, IMAP, LDAP, POP3, SMB, and SMTP.
- *tcpdump*
 - Packet and protocol analysis are essential parts of security.
 - *tcpdump* can be used to capture the packets from an interface and generate the *.pcap* file.
 - Can filter packets to save only a selected number of frames.
- *wireshark*
 - Packet capture and analysis are done using a graphical interface.
 - Captured packets can be easily analysed using *wireshark*.
- *hping*
 - *hping* is an open source packet generator and analyser for TCP/IP protocol.
 - It can be used to send large volumes of TCP traffic at a target while spoofing the source IP address to appear from a random or specified user source.
 - Subset of uses of *hping*:
 - Firewall testing
 - Advanced port scanning
 - Remote uptime guessing
 - TCP/IP stack auditing
- *tcpreplay*
 - Replays the previously captured traffic, which is saved in a *.pcap* file in the network interface.
- *lsattr*
 - Displays the information on the attributes of the specific device or type of the device.
- *dig*
 - This is the abbreviation for 'domain information groper'.
 - Like *nslookup*, it is used for querying the information about the domain from the DNS. The difference is that *nslookup* can be used for one server whereas *dig* can return the results of multiple servers at the same time.
- *lsdf*
 - Returns the list of open files and the processes that have opened them.
 - This command is used in UNIX-like operating systems.
- *dd*
 - Allows copying raw data from one source disk to another.
 - Called as 'disk/data duplicator'.
- *openssl*
 - OpenSSL is an open source command line tool that is commonly used to generate private keys, create CSRs, install your SSL/TLS certificate, and identify certificate information.
- *memdump*
 - Displays a hexadecimal memory dump defined by a starting address and specified number of bytes.
- *free*
 - Displays the total amount of free space available, along with the amount of memory used and swap memory in the system, and also the buffers used by the kernel.
- *ssh*
 - Provides a secure encrypted connection between two hosts over an insecure network.
- *nbtstat*
 - Displays NetBIOS over TCP/IP (NetBT) protocol statistics, NetBIOS name tables for both the local computer and remote computers, and the NetBIOS name cache.
 - Helps troubleshoot NetBIOS name resolution issues.
 - Displays the name of the current host system, and only root user authority can set the host name.
- *top*
 - Shows the real-time view of the processes running on the system.
- *systeminfo* / *uname*
 - Displays the information about the host, such as processor, kernel details, and operating system details.

Tools

Here are a few commonly used tools for enhancing security in digital networks.